

Practice Problem Set 2.1

CSE251 - Electronic Devices and Circuits

OPEN LOOP ANALYSIS OF OP-AMP

Voltage Level Detector, Square Wave Design, and
VTC Plots

[Course Description, COs,
and Policies](#)



[Midterm and Final
Questions](#)

Problem 1

- I. Determine the voltage input to the inverting terminal of an ideal op amp when $-40 \mu V$ is applied to the non-inverting terminal and the output through an open loop gain of 150,000 is 15 V.
- II. Determine the voltage across the $10 \text{ k}\Omega$ resistor for the circuit shown below. Assume the op-amp to be ideal except with a finite gain $A = 10^4$.

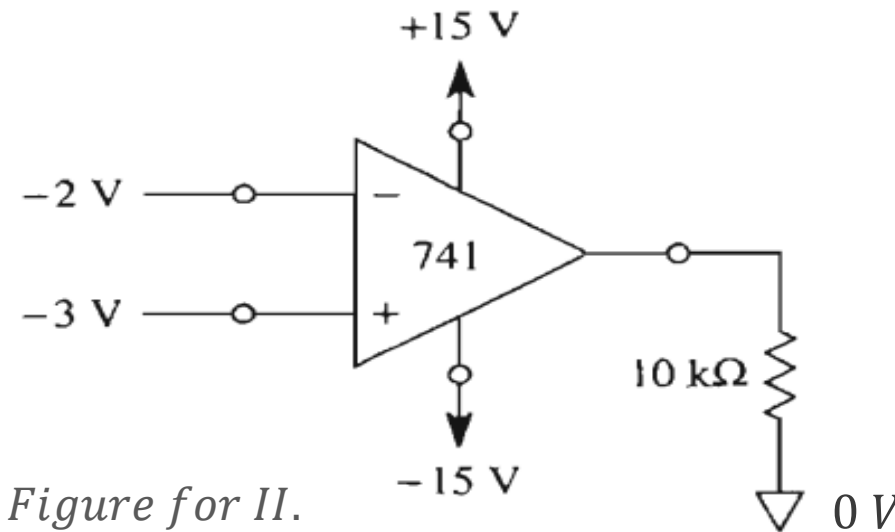
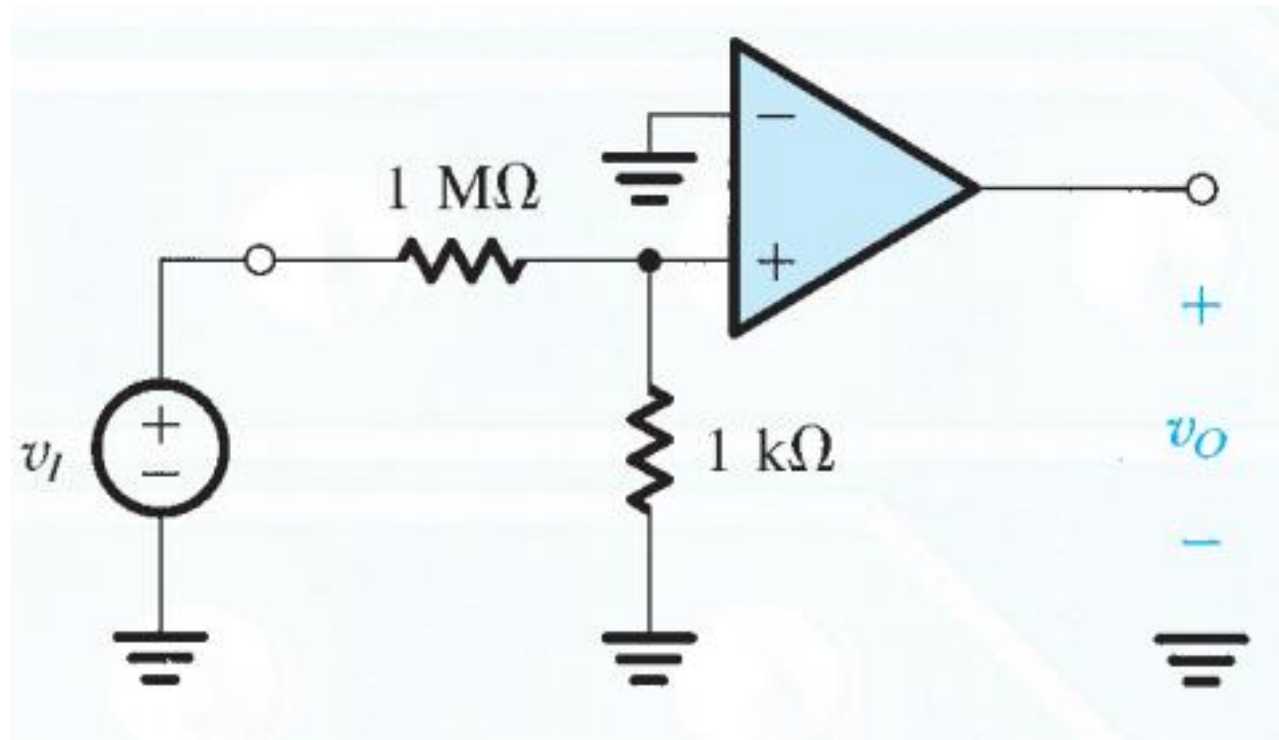


Figure for II.

Ans: I. $-140 \mu V$; II. -15 V

Problem 2

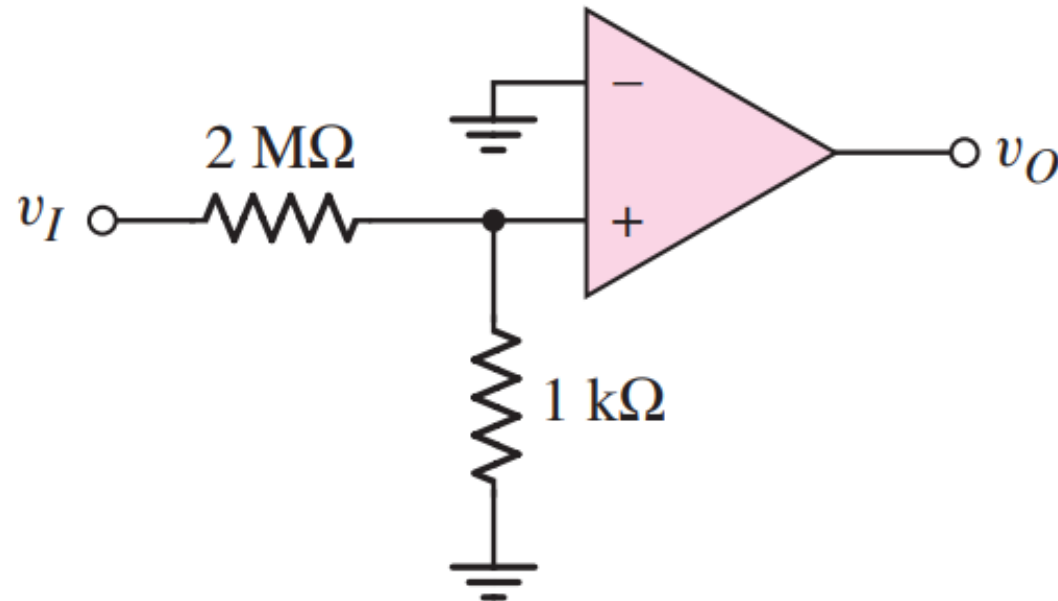
- The following circuit uses an op amp that is ideal except for having a finite gain A . Measurements indicate $v_O = 4.0\text{ V}$ when $v_I = 2.0\text{ V}$. What is the op-amp gain A ?



Ans: $A = 2002$

Problem 3

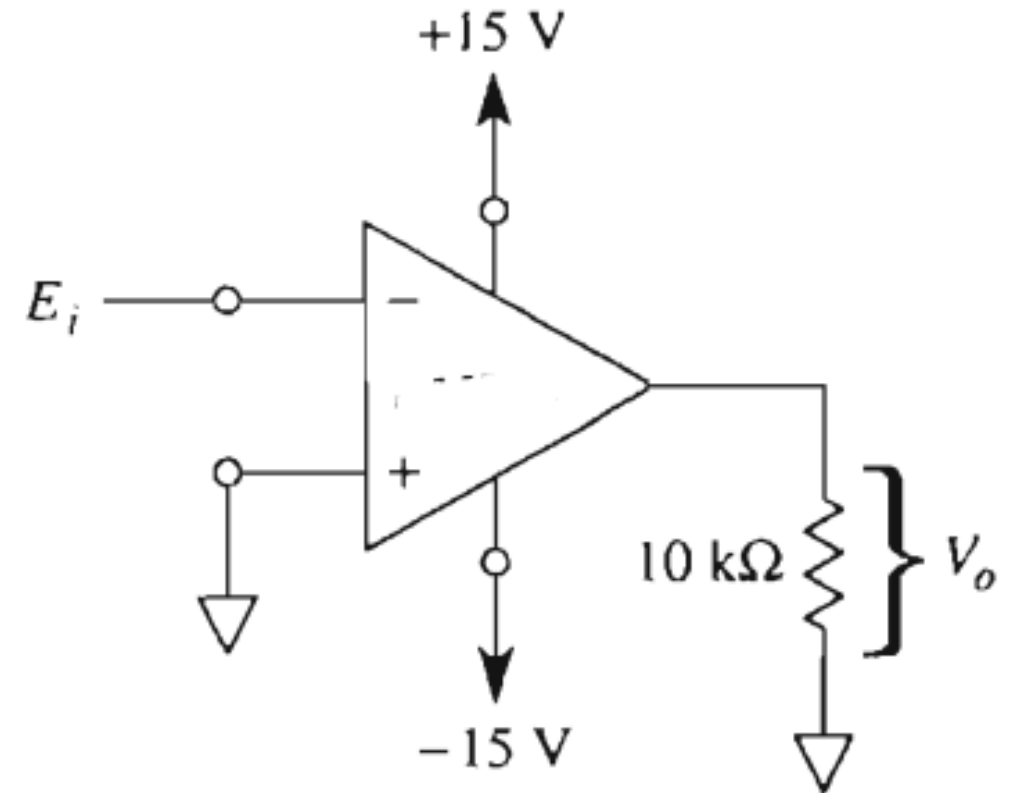
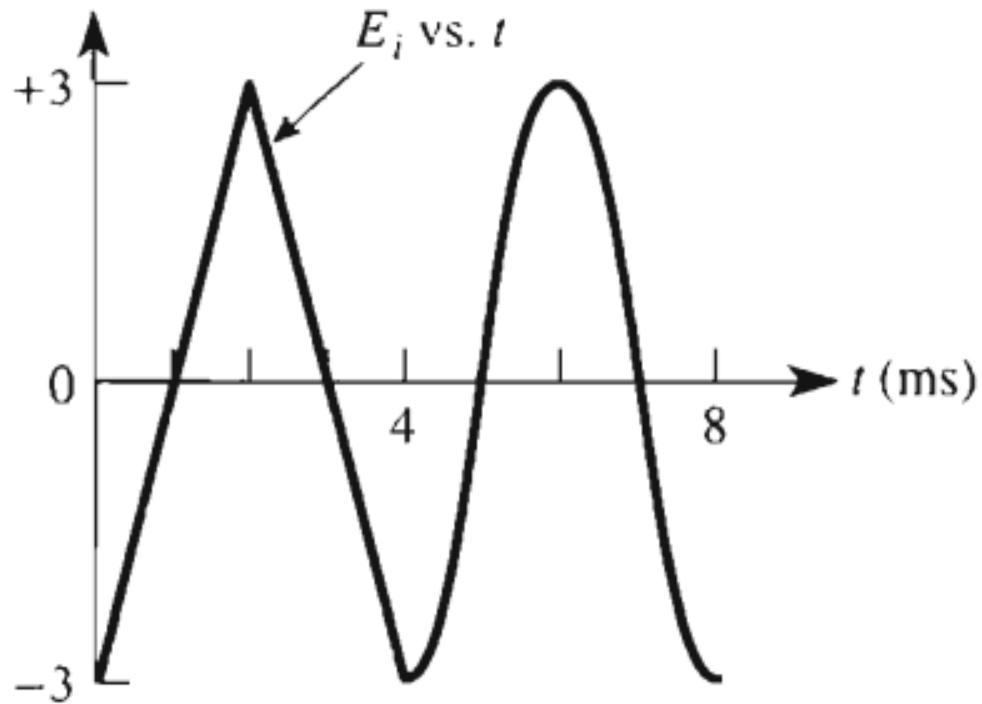
- The following circuit uses an op amp that is ideal except for having a finite gain A . If $A = 10^4$ and $v_O = -2.0\text{ V}$, determine v_I .



Ans: $v_I = -0.4002\text{ V}$

Problem 4

- Sketch V_o vs. t and V_o vs. E_i .

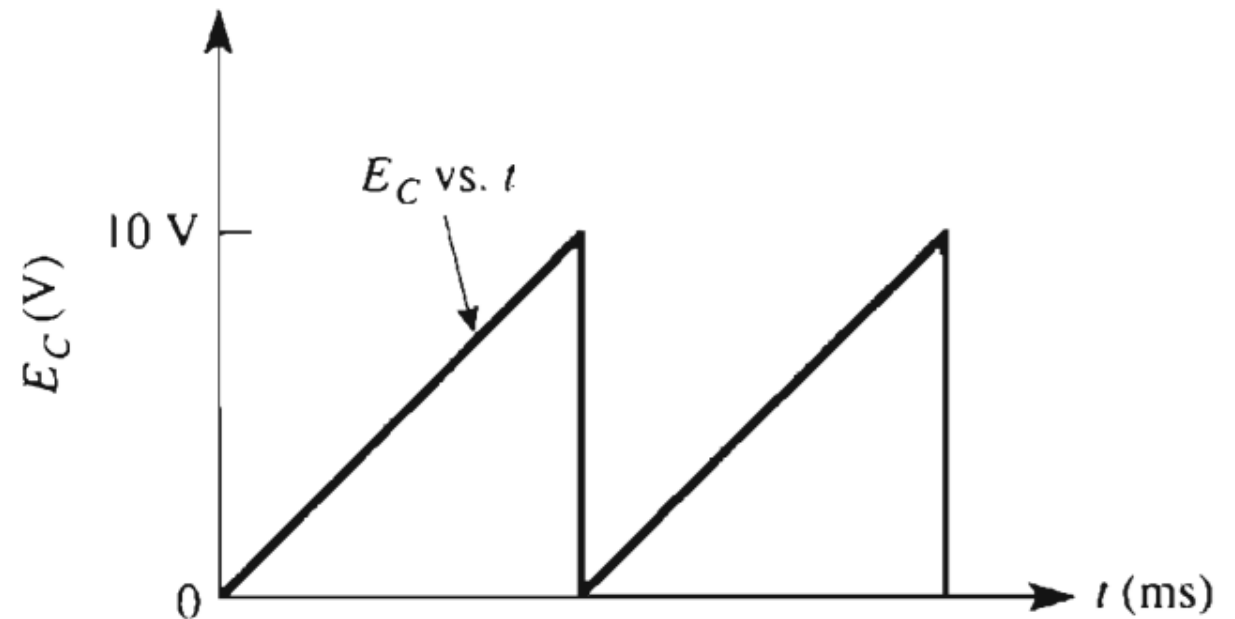
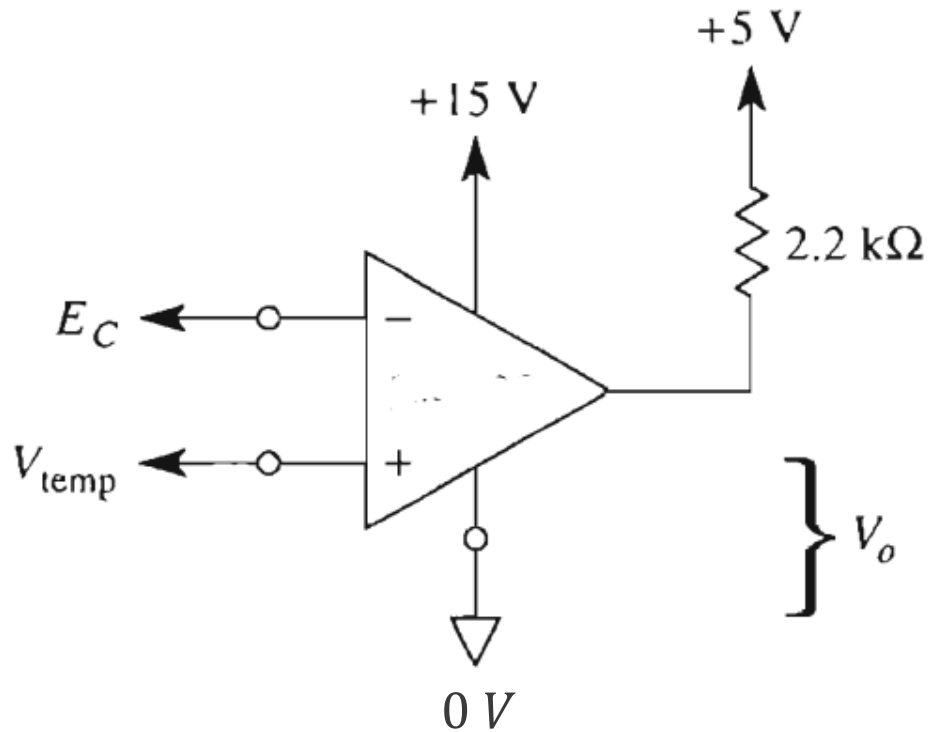


Problem 5

- I. Draw the schematics of a circuit whose output voltage will go positive to $+15\text{ V}$, when the input signal crosses $+5\text{ V}$ in the positive direction.
- II. Draw the schematics of a circuit whose output voltage will go positive to $+15\text{ V}$, when the input signal is below -4 V . The output should be at -15 V when the input is above -4 V .

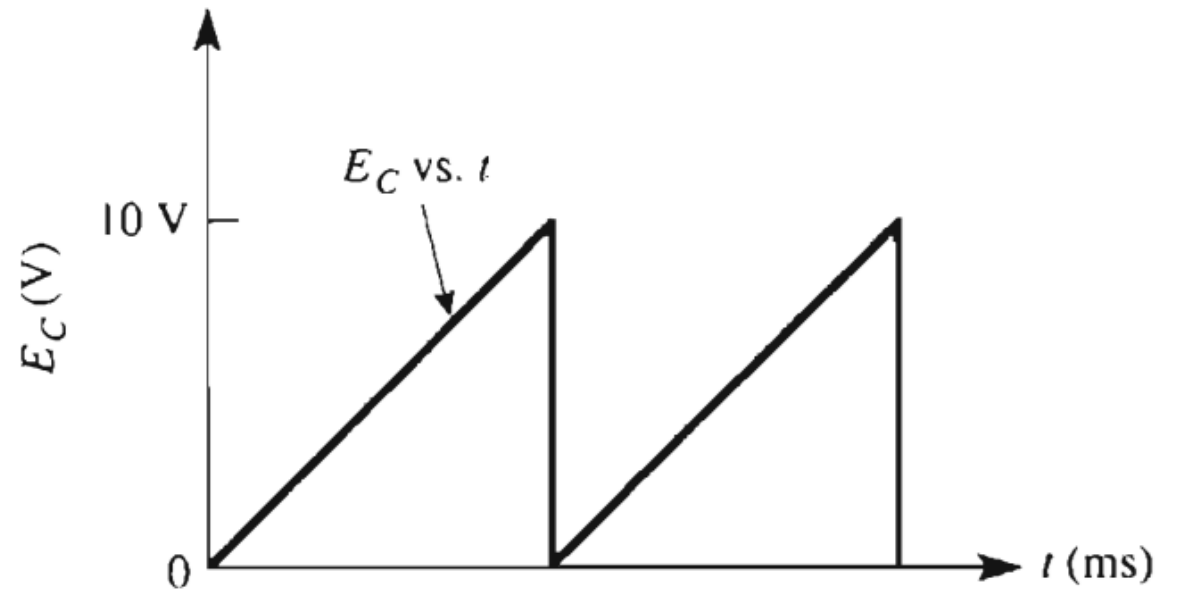
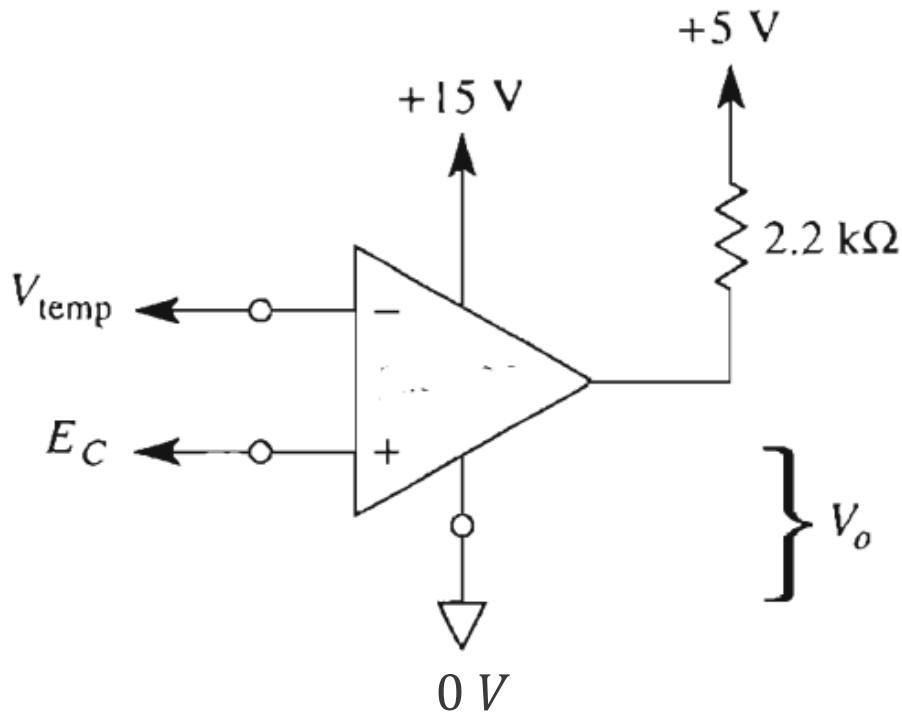
Problem 6

- The frequency of E_C is constant at 50 Hz. If $V_{temp} = 4\text{ V}$, (a) plot V_o vs. E_C , (b) plot V_o vs. t , and (c)** calculate the high time of V_o .



Problem 7

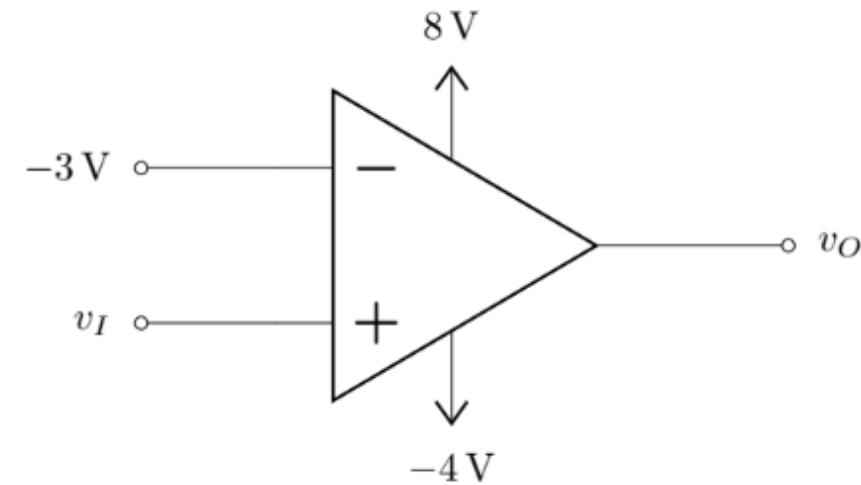
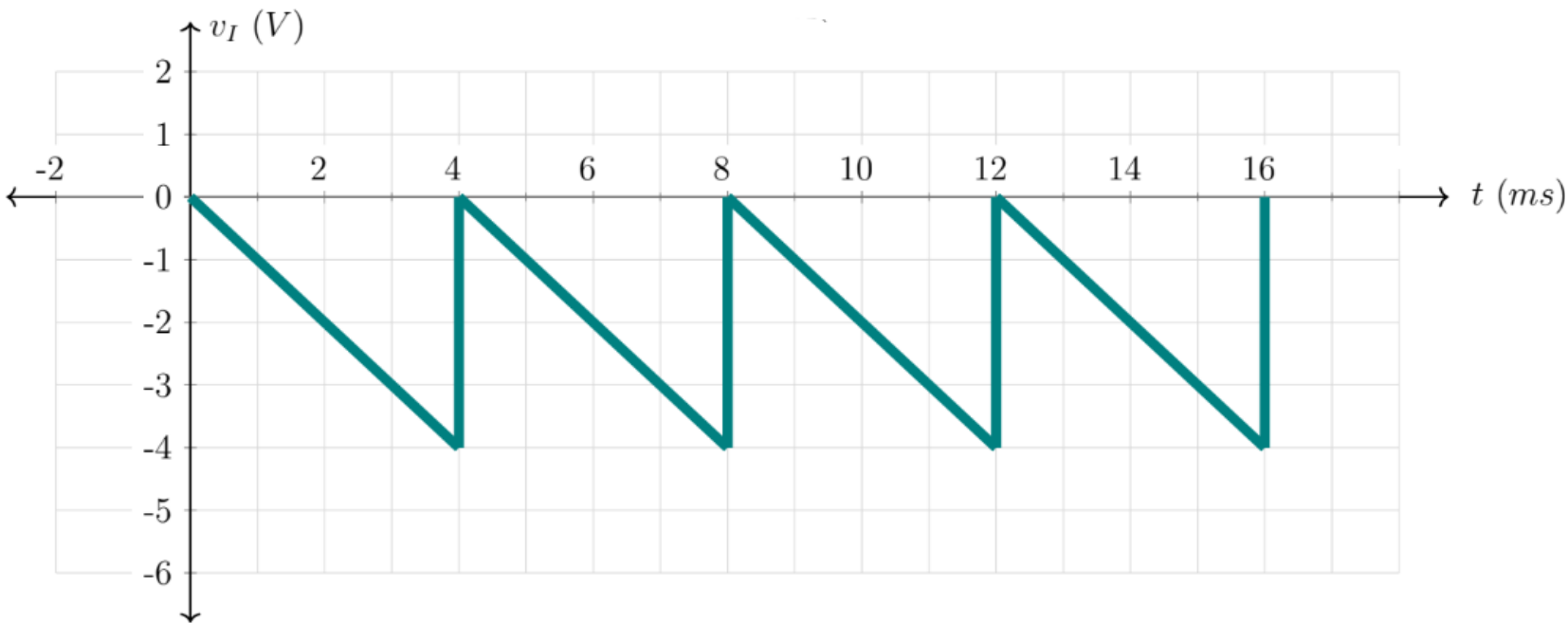
- The frequency of E_C is constant at 50 Hz. If $V_{temp} = 3\text{ V}$, (a) plot V_o vs. E_C , (b) plot V_o vs. t , and (c)** calculate the high time of V_o .



Note: Problems marked with an asterisk (**) are a bit more advanced for this course. However, attempting them can help you develop a stronger grasp of the topic.

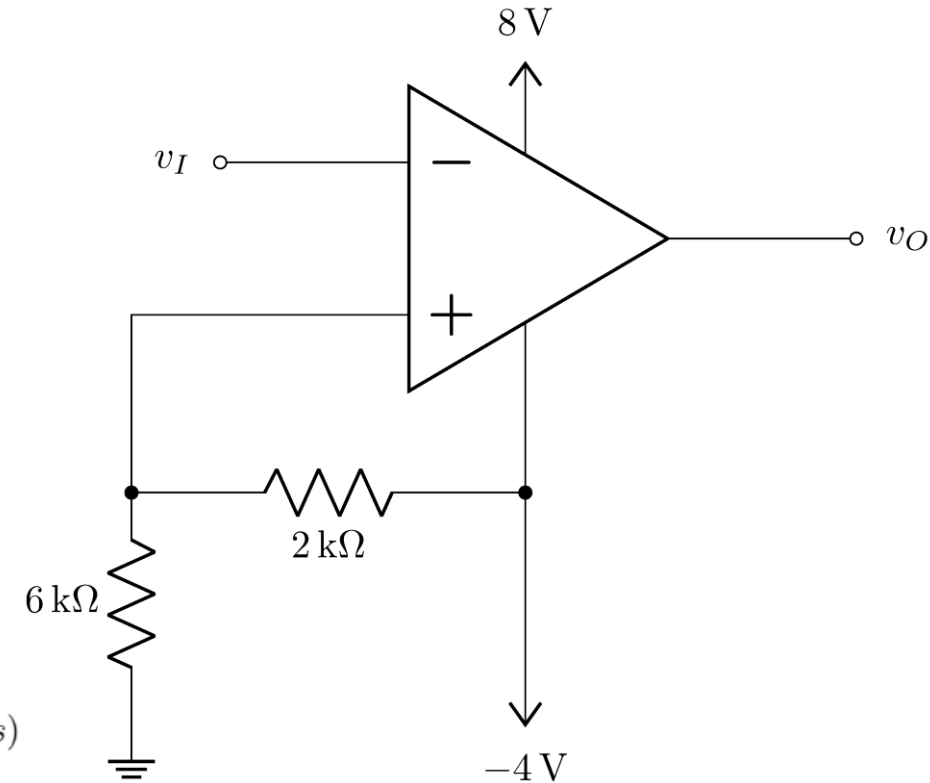
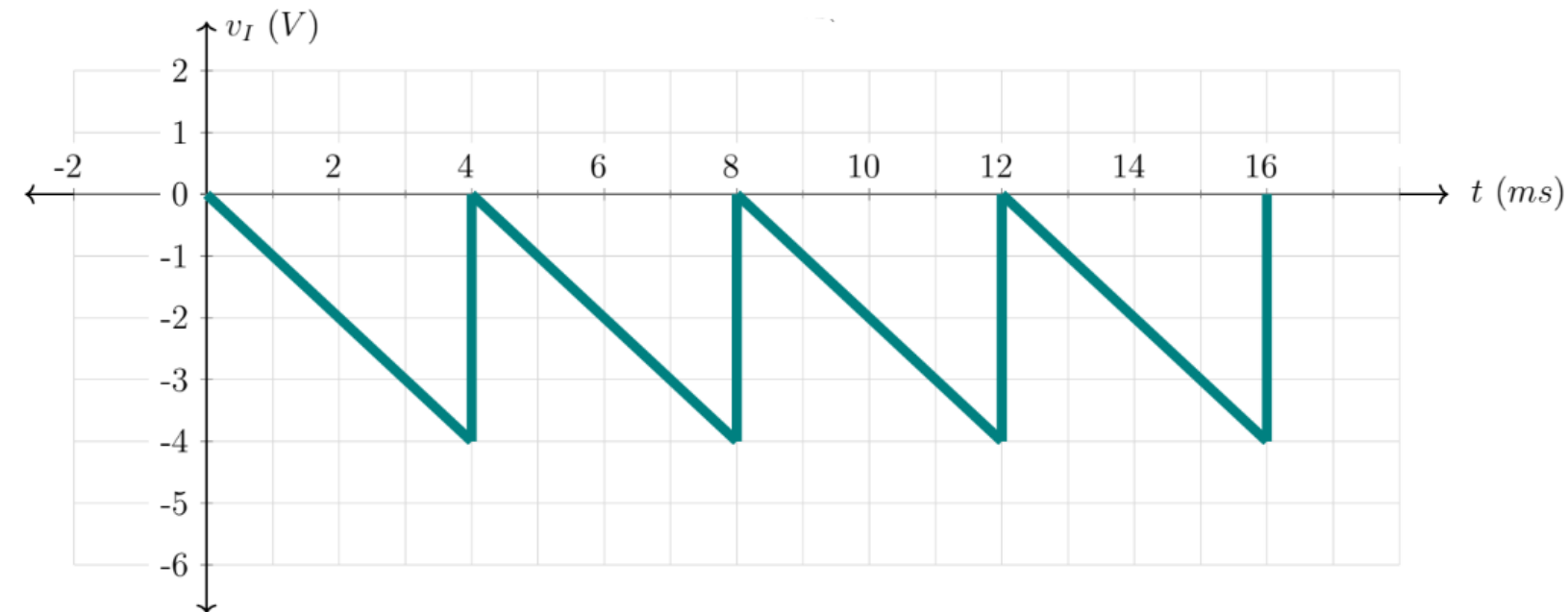
Problem 8

- In the adjacent circuit, v_I is a time varying voltage as plotted below. Sketch the graphs of (a) V_o vs. t and (b) V_o vs. v_I .



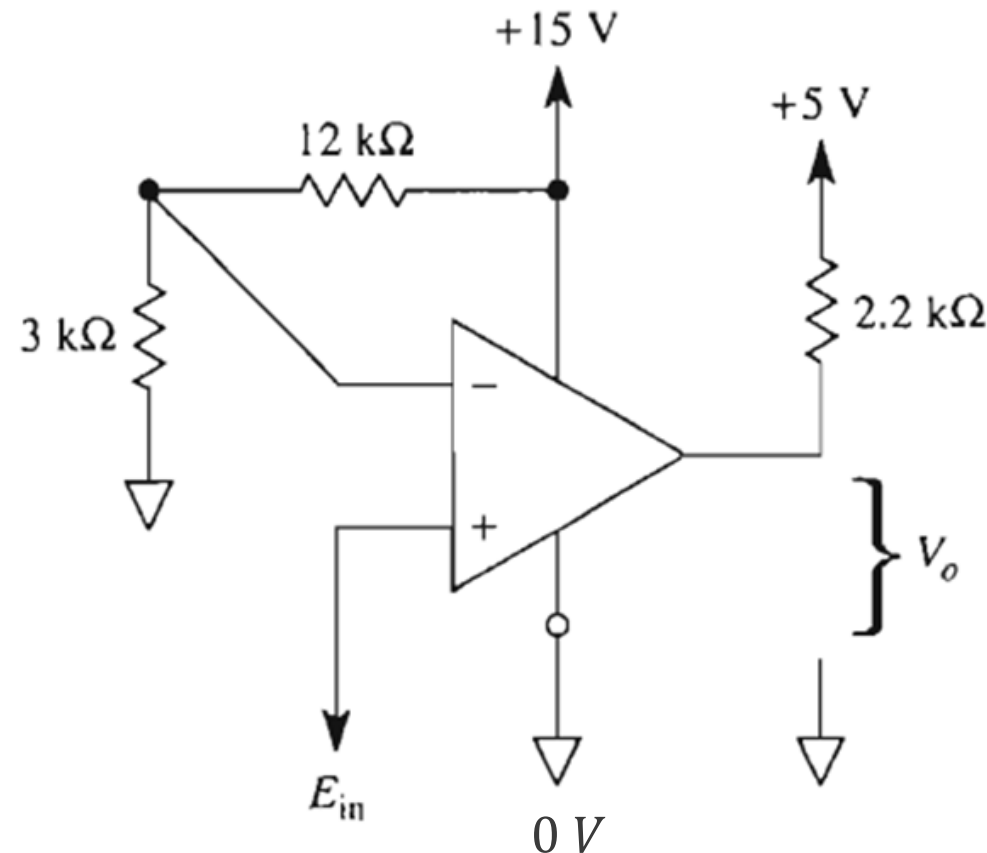
Problem 9

- In the following circuit, v_I is a triangular wave as plotted below. Sketch the graphs of (a) V_o vs. E_{in} and (b) V_o vs. t .



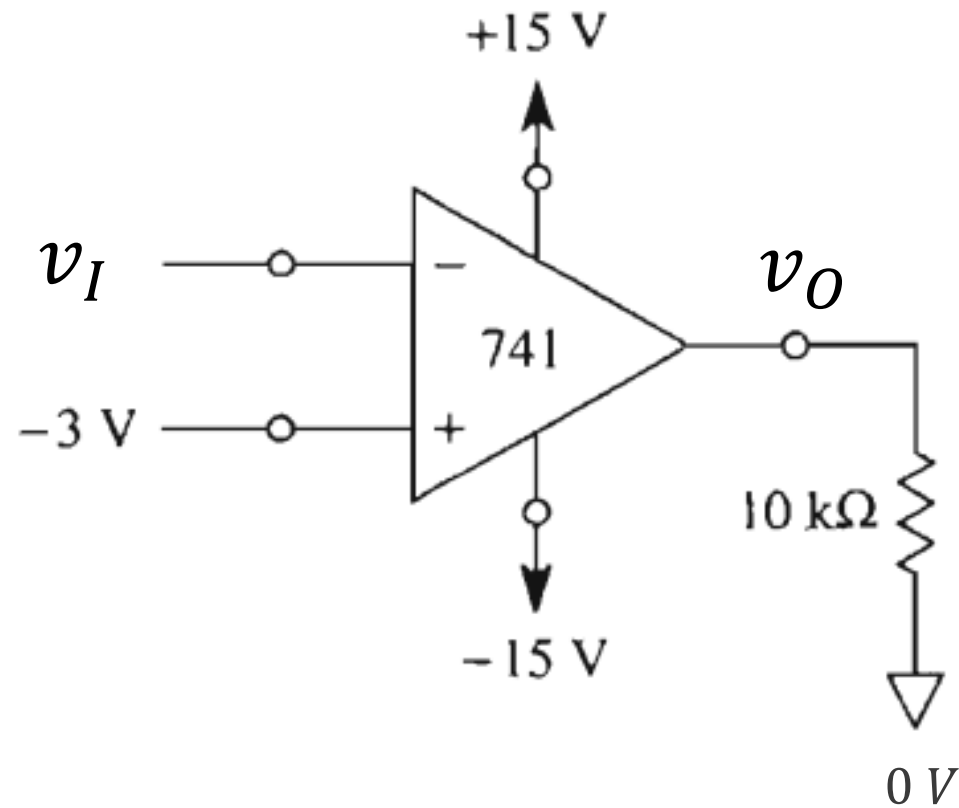
Problem 10

- In the following circuit, E_{in} is a triangular wave of 100 Hz frequency with an amplitude of 5 V. Sketch, with appropriate levels of axes crossings, the graphs of (a) V_o vs. E_{in} and (b) V_o vs. t .



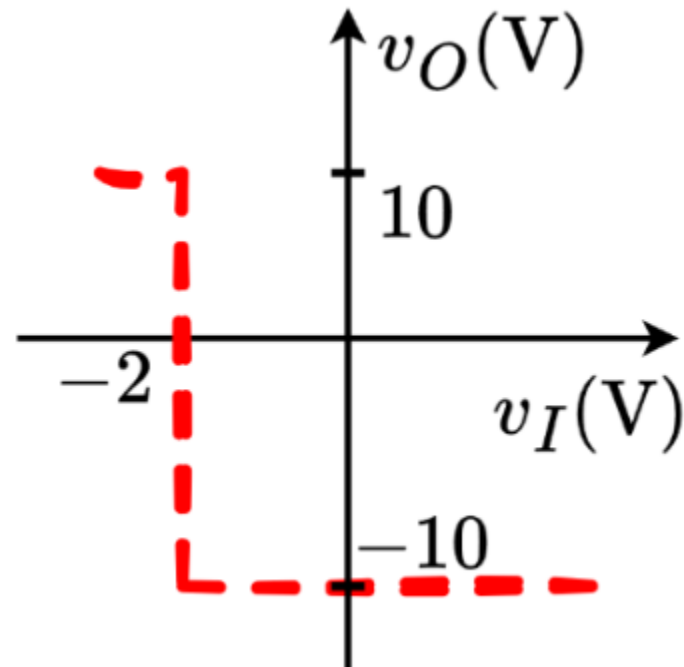
Problem 11

- Draw the Voltage Transfer Characteristics (v_O vs. v_I) of the following circuit.



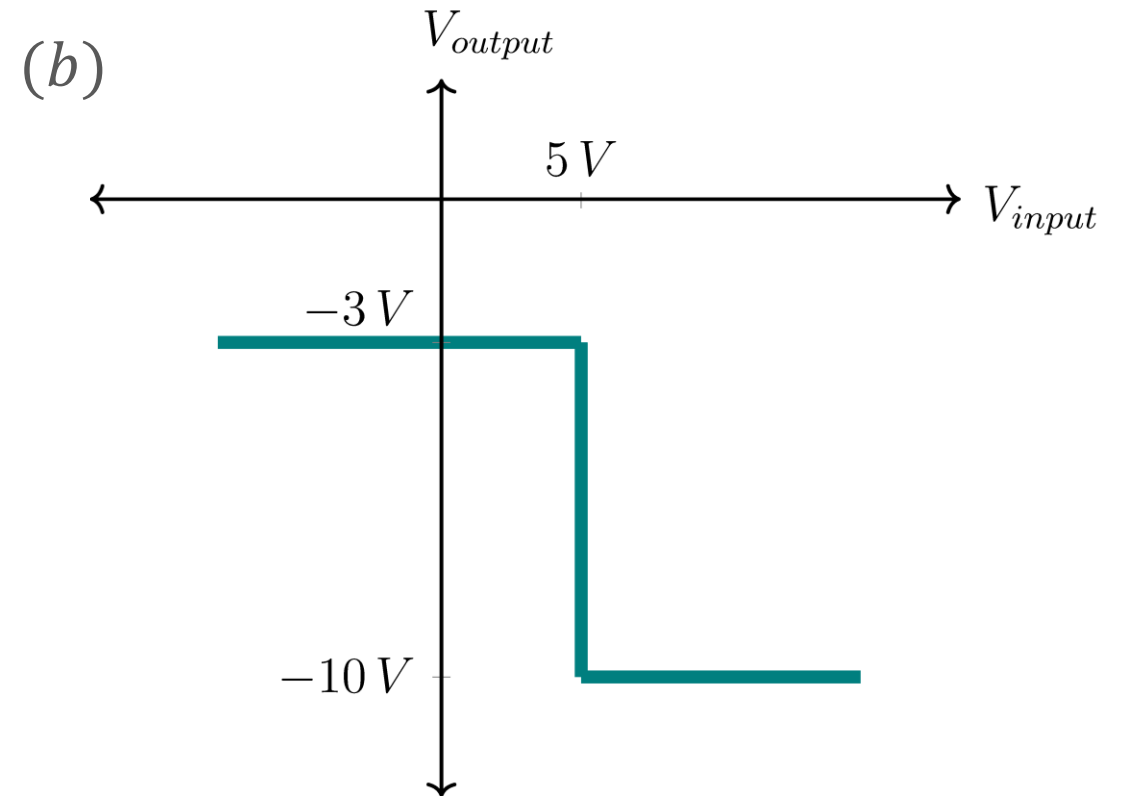
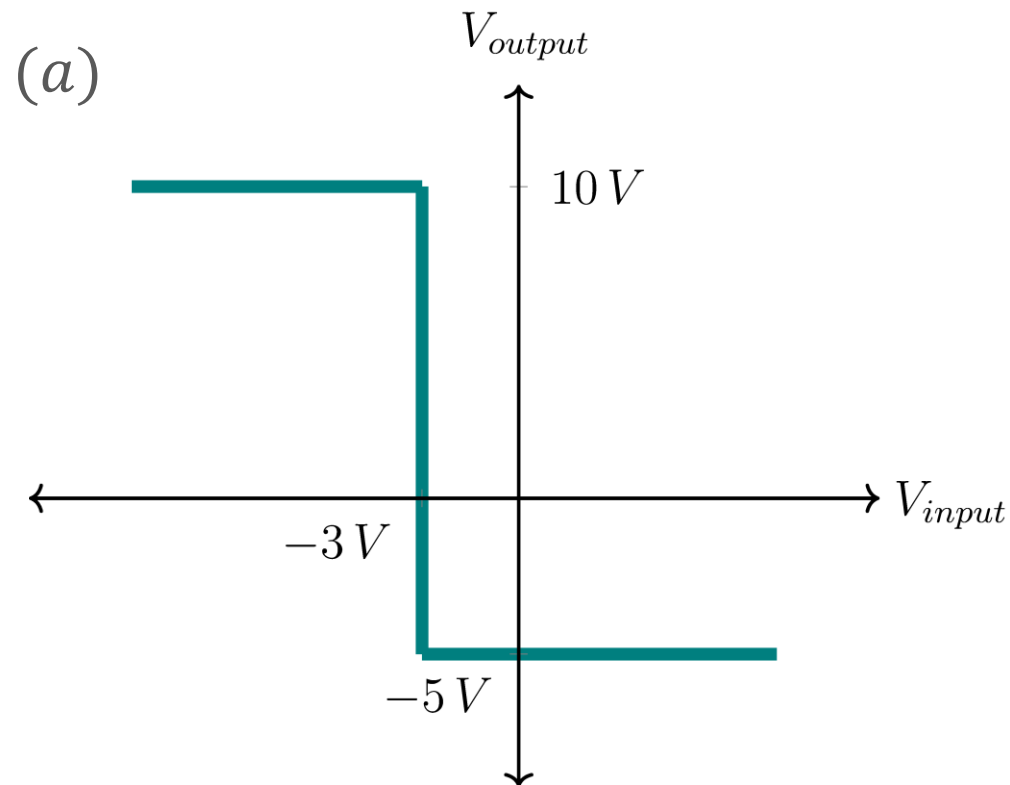
Problem 12

- Design a circuit with a single ideal op-amp for each of the VTC plots shown below. v_O and v_I are the output voltage and the input voltage respectively.



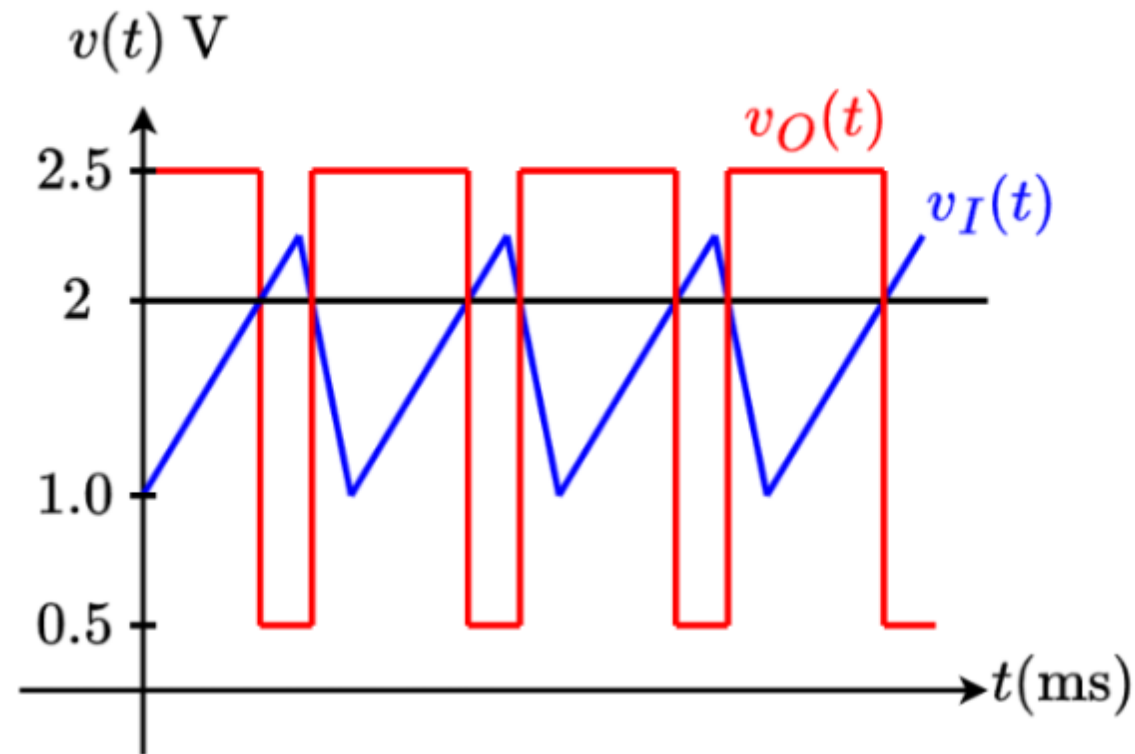
Problem 13

- Design a circuit with a single ideal op-amp for each of the VTC plots shown below. V_{output} and V_{input} are the output voltage and the input voltage respectively.



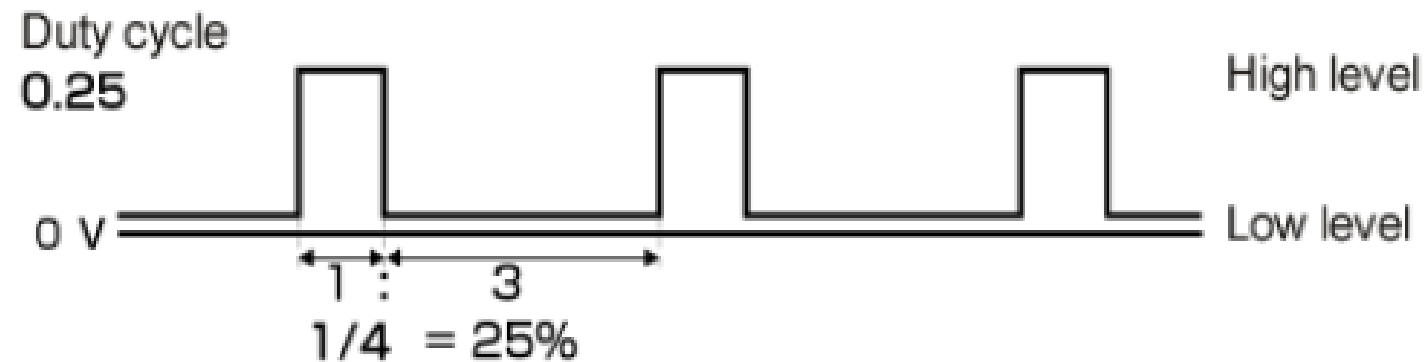
Problem 14

- I. Draw the Voltage Transfer Characteristics (v_O vs. v_I) of the following circuit.
- II. Also design a circuit that would give rise to the VTC you plotted in I.



Problem 15**

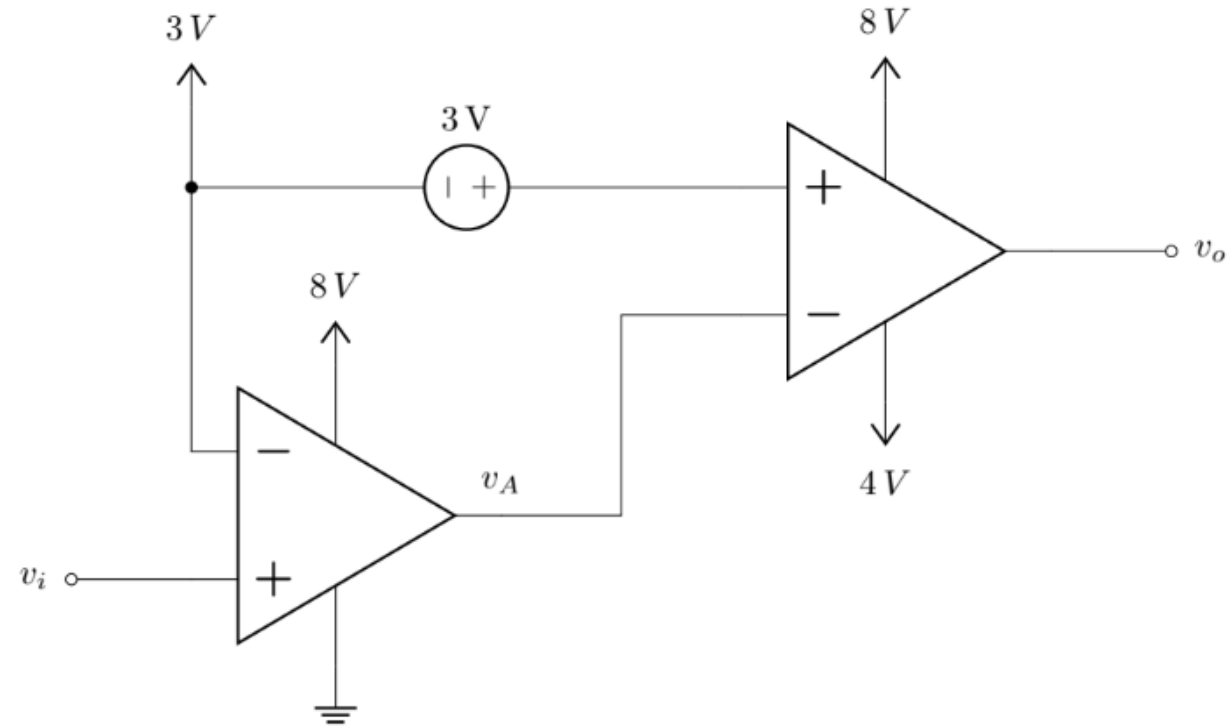
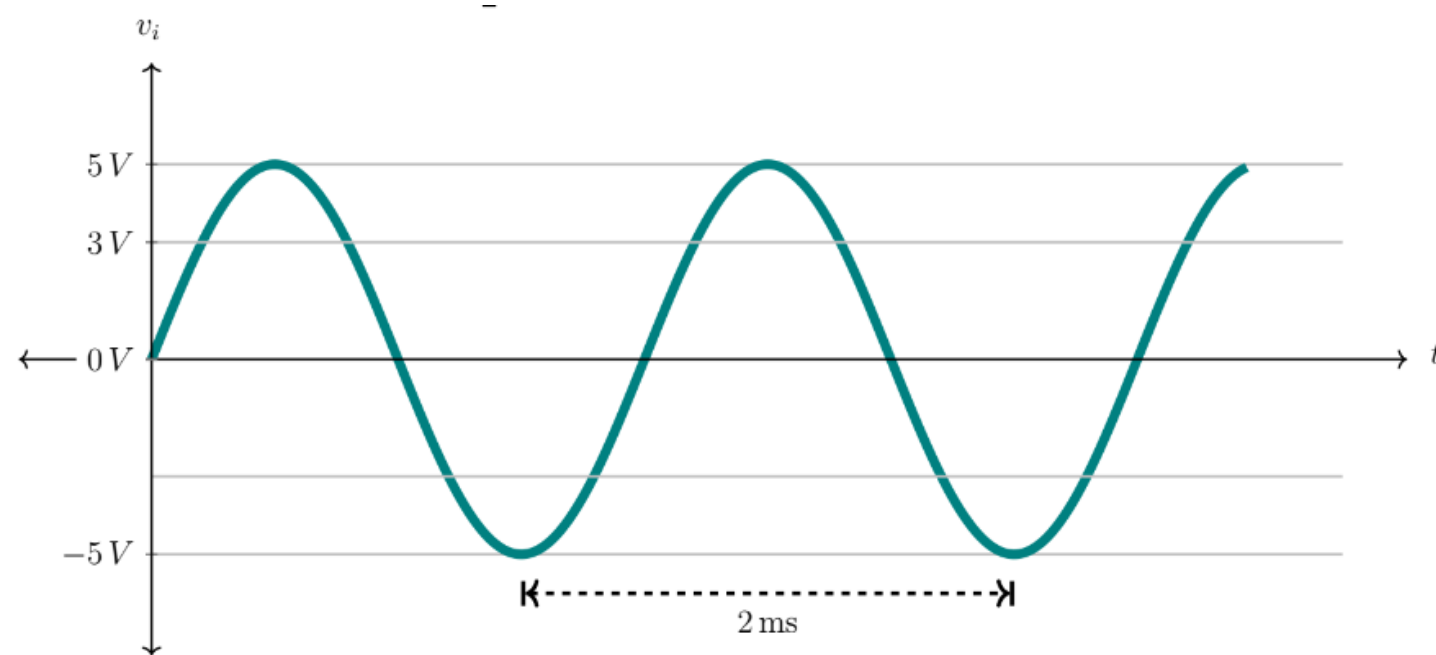
- Design a circuit using op-amp to transform the sinusoidal voltage $v_I = 6 \sin(200\pi t)$ (V) into a square wave with 25% duty cycle as shown below.



Note: Problems marked with an asterisk (**) are a bit more advanced for this course. However, attempting them can help you develop a stronger grasp of the topic.

Problem 16

- Sketch v_A vs. t and v_o vs. t .



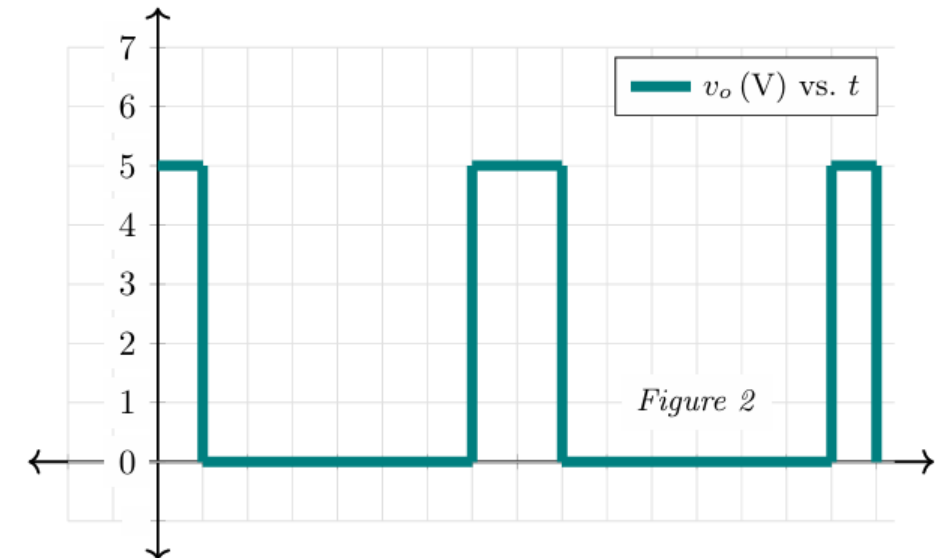
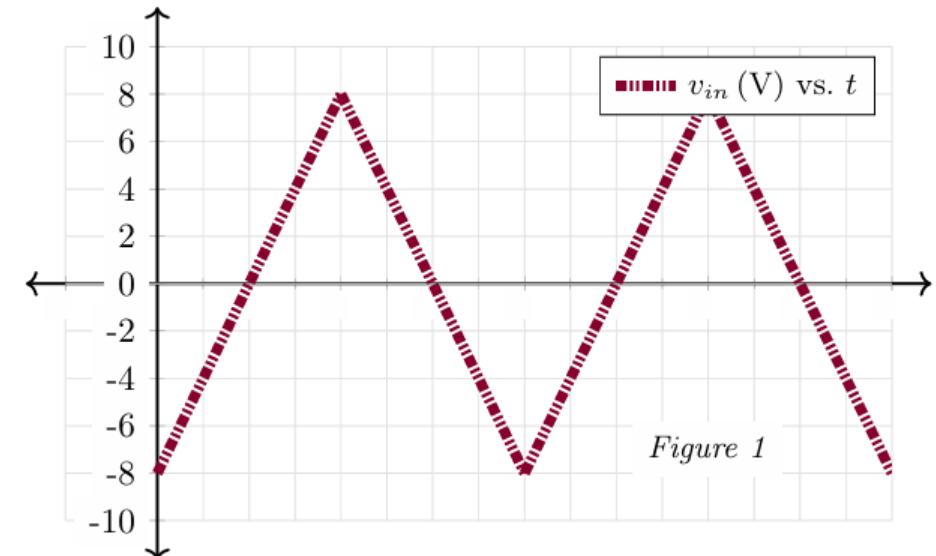
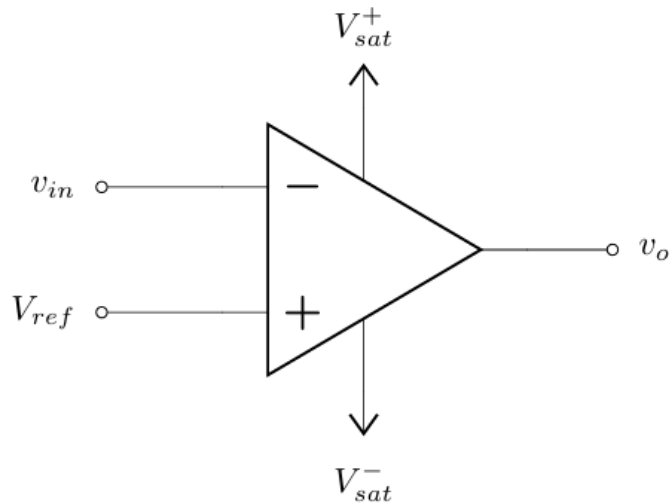
Up Next

Relevant *Questions* from Past Semesters

Mid — Fall 2025

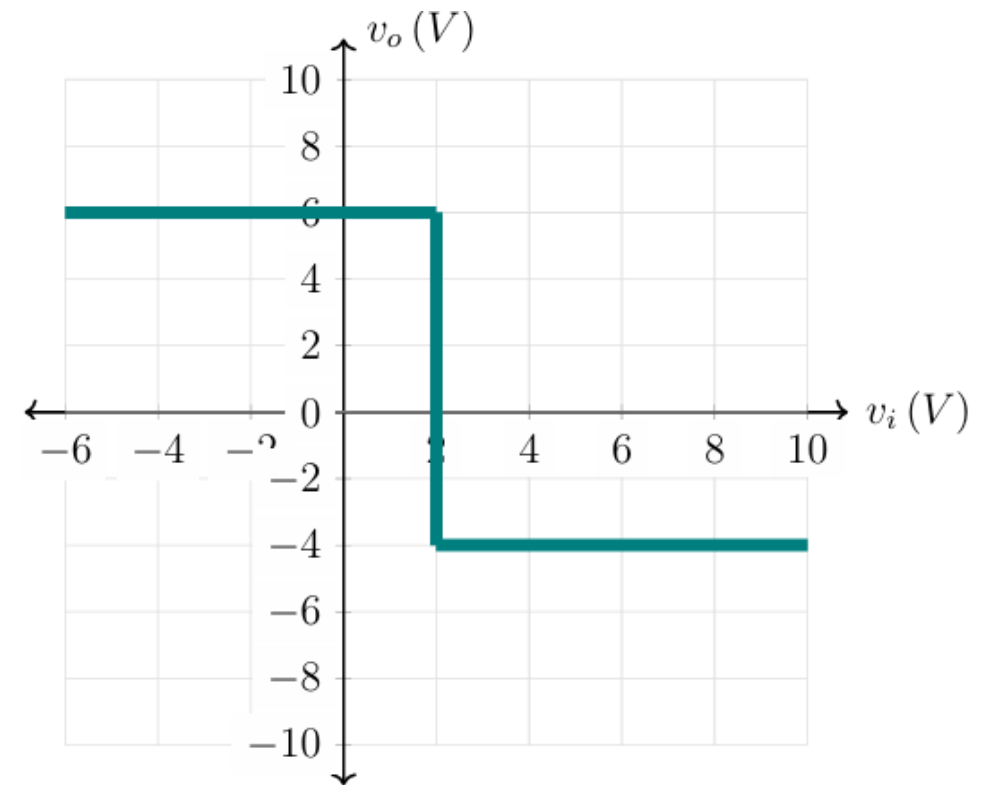
- In the following circuit, v_{in} is a triangular voltage plotted as a function of time t in *Figure 1*, and V_{ref} is a constant DC voltage. The output voltage waveform v_o is shown on *Figure 2*.

Determine V_{ref} and the VTC of the circuit.



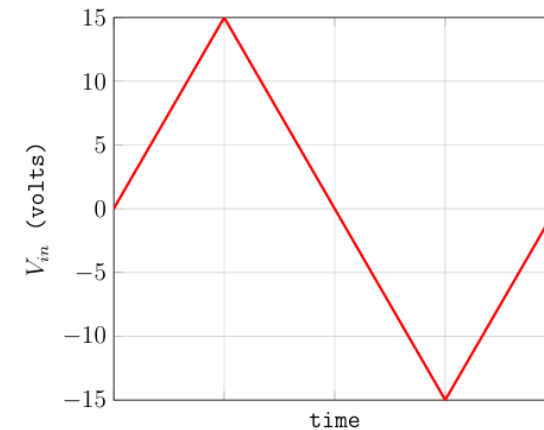
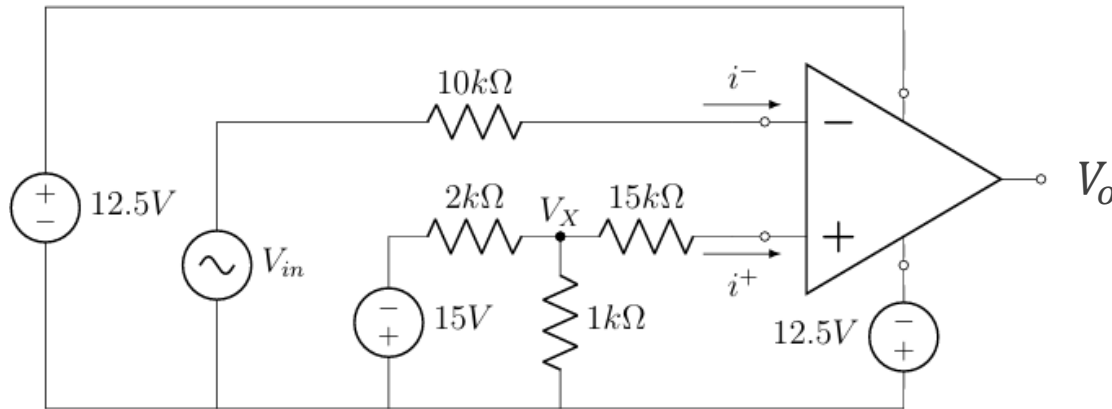
Mid — Summer 2025

- Design an op-amp circuit that follows the VTC shown below. v_i and v_o represent the input and output voltages, respectively.

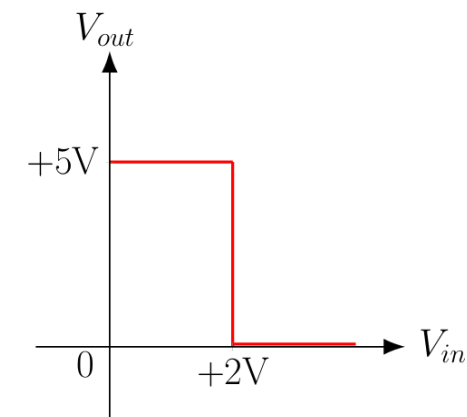


Mid — Spring 2025

a. If V_{in} is as shown in the following plot, sketch V_o as a function of time.

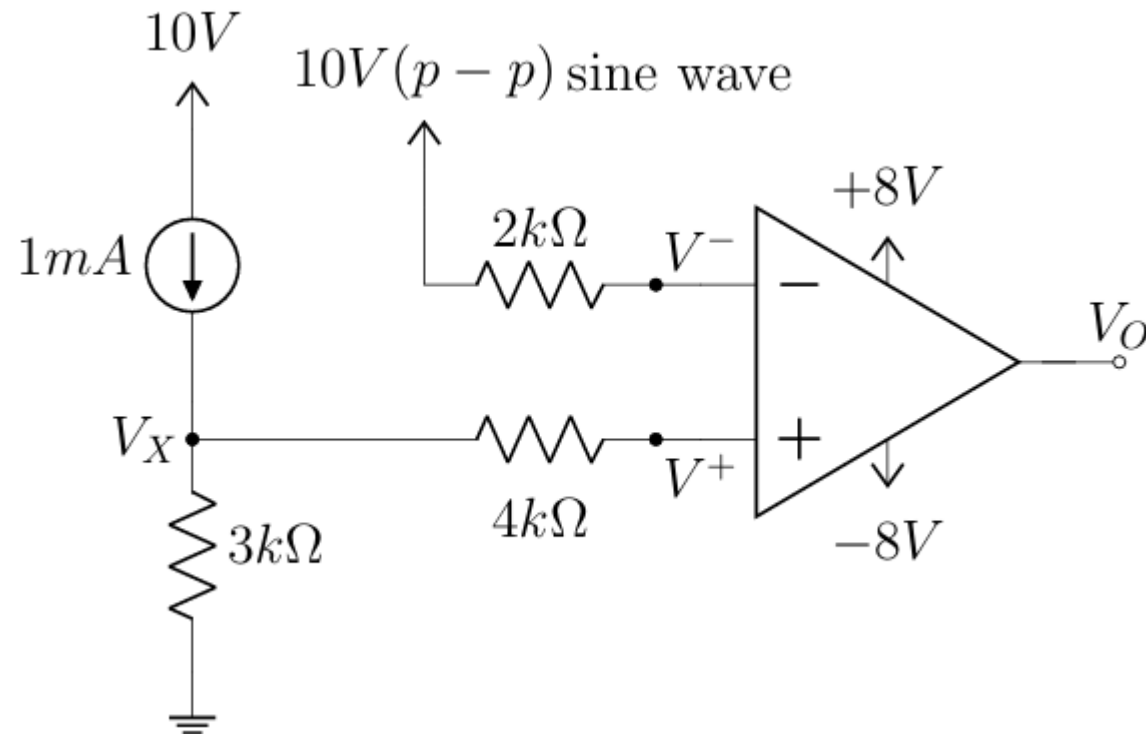


b. The VTC of an ideal NOT gate can be represented by the adjacent graph. Design a circuit using an op-amp comparator that can work as an ideal NOT gate.



Mid — Fall 2024

- Draw the waveform of V_O in the circuit shown below.



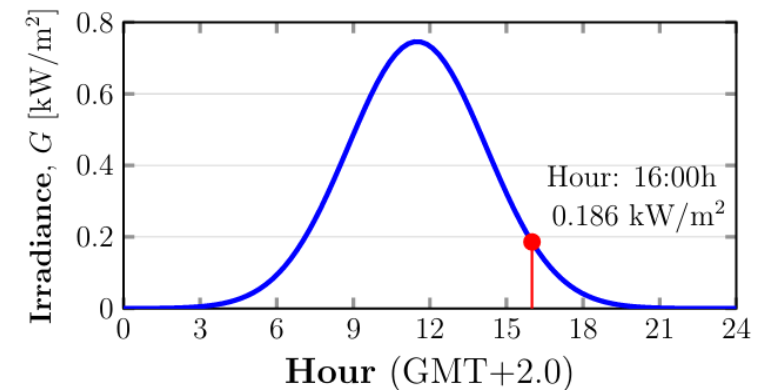
Mid — Spring 2024

The following figure shows the variation of direct solar irradiance, G (in units of kW/m^2) throughout the day in Egypt. The irradiance is $0.186 \text{ kW}/\text{m}^2$ at 16:00 hours. A photodetector generates a voltage of V_S by converting solar irradiance G according to the following equation:

$$V_S(G) = 6G^2 + 7G$$

You have to design an automatic system with an Op-Amp comparator circuit. This system will take V_S as an input from a photodetector and control a street lamp, ensuring it meets the following conditions:

- The lamp switches ON after 16:00 hours.
- The lamp switches ON when system output is 5 V.
- The lamp switches OFF when system output is 0 V.



Draw the completed Op-Amp comparator circuit clearly marking the input and output terminals and the positive and negative saturation voltages. Also clearly indicate the reference voltage against which the input of the comparator is compared.

Final — Spring 2024

BRAC University has installed Air Conditioning systems in its classrooms. The system monitors the signals from two temperature sensors T_1 and T_2 . A temperature sensor is supposed to turn *HIGH* if the temperature of a classroom exceeds 25°Celsius . The system uses the signals from two sensors so that it is more reliable.

However, the implemented AC system is found defective. The AC turns *ON* if any one of the sensors turn *HIGH*, but unfortunately, the AC turns *OFF* if both the sensors become *HIGH*. As a result, in some cases of extreme heat, the AC turns off which causes a lot of suffering to both teachers and students during class hours.

- Construct the logic circuit that the (defective) Air Conditioning system uses using MOSFETs.
- Propose a new logic circuit to resolve the defect.

Final — Summer 2022

Design a circuit using Op-Amp comparator to automatically turn *ON* and *OFF* the streetlights. For this, you have a lux sensor installed on top of the streetlights (facing above) that outputs a voltage proportional to amount of natural light, as listed below:

$v_{night, 0 \text{ lux}} = 1 \text{ V}$	$v_{dusk, 20 \text{ lux}} = 2 \text{ V}$	$v_{dawn, 80 \text{ lux}} = 3 \text{ V}$
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The lights require 20 V and should be *ON* if the amount of light goes below 20 lux (at dusk).

Acknowledgement and References

Some of the problems in this set are taken or adapted from the following sources:

1. Sedra, A. S., & Smith, K. C., Microelectronic Circuits, Oxford University Press
2. Coughlin, R. F., & Driscoll, F. F., Operational Amplifiers and Linear Integrated Circuits, Pearson
3. Neamen, D. A., Microelectronics: Circuit Analysis and Design, McGraw-Hill